

Practical 10 CC Lab

```
{
#include <stdio.h>

int yylex();
void yyerror(char *s);
%}

%token FOR ID NUM INC DEC

%<>
%<4>

program:
    for_loop
    {
        printf("Valid for loop\n");
    }
    ;

for_loop:
    FOR '(' initialization ';' condition ';' increment ')'
    ;

initialization:
    ID '=' NUM
    ;

condition:
    ID '<' NUM
    | ID '>' NUM
    ;

increment:
    ID INC
    | ID DEC
    ;

%<>
%<0>

void yyerror(char *s)
{
    printf("Invalid for loop\n");
}

int main()
{
    printf("Enter a for loop:\n");
    yyparse();
    return 0;
}
```

```
lab-03-28@lab-03-28-OptiPlex-3280-AIO:~$ vim exp10_275.l
lab-03-28@lab-03-28-OptiPlex-3280-AIO:~$ vim exp10_275.y
lab-03-28@lab-03-28-OptiPlex-3280-AIO:~$ yacc -d exp10_275.y
lab-03-28@lab-03-28-OptiPlex-3280-AIO:~$ lex exp10_275.l
lab-03-28@lab-03-28-OptiPlex-3280-AIO:~$ gcc y.tab.c lex.yy.c -o exp10_275 -ll
lab-03-28@lab-03-28-OptiPlex-3280-AIO:~$ ./exp10_275
Enter a for loop:
for(i=0;i<10;i++)
Valid for loop
^C
lab-03-28@lab-03-28-OptiPlex-3280-AIO:~$ ./exp10_275
Enter a for loop:
for(i=0 i<10; i++)
Invalid for loop
lab-03-28@lab-03-28-OptiPlex-3280-AIO:~$
```

Practical 10 CC Lab

```
%{
#include "y.tab.h"
%}

%%
"for"      { return FOR; }
"++"       { return INC; }
"--"       { return DEC; }

[a-zA-Z][a-zA-Z0-9]* { return ID; }
[0-9]+      { return NUM; }

"="        { return '='; }
"<"        { return '<'; }
">"        { return '>'; }
";"        { return ';'; }
"("        { return '('; }
")"        { return ')'; }

[ \t\n]    ;

.          { return yytext[0]; }
%%

int yywrap()
{
    return 1;
}
```